

Ahmed Baig

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SUMMARY

Full-stack engineer with 3 years of experience building end-to-end web applications, pairing React/Next.js frontends with scalable Node.js/NestJS and Python backends on PostgreSQL. Experienced across the full lifecycle: designing REST APIs, integrating third-party services (Twilio, AWS, Google Gemini), building responsive UIs, and shipping with Docker and CI/CD. Hands-on experience delivering LLM-powered features and automation workflows including tool/function calling and prompt engineering.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, HTML/CSS

Frontend: React, Next.js, React Query, WebSockets, responsive UI, component design

Backend & APIs: Node.js, NestJS, Express.js, Flask, REST API design, GraphQL, Swagger/OpenAPI

Databases: PostgreSQL, PostGIS, Redis, MySQL, MongoDB, DynamoDB; ORMs: Sequelize, TypeORM

API & Systems: Authentication (JWT), caching (Redis), rate limiting, pagination, automated testing (Jest), performance optimization

AI & Automation: LLM API integration (Google Gemini), tool/function calling, prompt engineering, structured output, n8n workflow automation

Cloud & DevOps: AWS (Lambda, Cognito, Amplify, DynamoDB, S3), Docker, GitHub Actions, Jenkins, Terraform

EDUCATION

National University of Computer and Emerging Sciences

Bachelor of Science in Computer Science

Islamabad, Pakistan

2020 – 2024

EXPERIENCE

Backend Specialist

BaKhabar Kissan (BKK)

June 2025 – Present

Islamabad, Pakistan

- Developing robust, high-performance backend systems and APIs using Node.js, NestJS, and Python, plus delivering frontend for 3 portals in React/Next.js from design to deployment.
- Designing and implementing end-to-end database and system flow architectures on PostgreSQL, laying the foundation for scalable and maintainable applications.
- Engineering and maintaining complex ETL pipelines, ensuring reliable and efficient data extraction, transformation, and loading across multiple systems.
- Leading deep-dive optimizations of relational database performance through profiling and query tuning, specializing in advanced SQL and PostGIS (Materialized Views, spatial indexing, complex Joins).
- Driving ecosystem-wide performance optimization across 9 projects; spearheading complete end-to-end backend architecture for 5 key projects.

Full Stack Engineer

QLU.ai

Aug 2024 – May 2025

Islamabad, Pakistan

- Developed and enhanced production-level features across the stack using NestJS, React, React Query, and PostgreSQL, ensuring scalable and efficient web applications.
- Designed and implemented communication systems (incoming/outgoing calls, messaging) by integrating Twilio's third-party API, including callback/webhook handling for real-time events.
- Built and expanded core UI functionalities such as anonymous contact handling, contact notes, and real-time messaging, significantly improving user experience and workflow efficiency.
- Optimized API performance and data processing, enabling efficient handling of 80,000+ data items in a single API call through algorithmic improvements and database optimizations.
- Developed a **real-time chat application** using WebSockets and the PERN stack, ensuring low-latency communication and robust performance.

Fellowship Associate - Backend

Fasset

June 2024 – Aug 2024

Islamabad, Pakistan

- Contributed to developing a **Flexible Transaction Fee System**, implementing fee calculation logic, JWT authentication, Redis caching, and input validation using Sequelize ORM (PostgreSQL).

- Gained expertise in MVC backend architecture while collaborating with Fasset’s national and international teams during the fellowship.

Teaching Assistant (CS4032: Web Programming)

Feb 2024 – Jun 2024

National University of Computer and Emerging Sciences

Islamabad, Pakistan

- Conducted interactive demo sessions on core web development concepts, including JavaScript, React.js, and Node.js, for a class of 60 students.

Backend Development Intern

July 2023 – Aug 2023

MRS Technologies

Islamabad, Pakistan

- Built the backend of an **IoT-based microgreens plant system** using AWS Amplify, DynamoDB, GraphQL, and AWS Lambda with Cognito sign-up. Selected as one of the top 15 interns out of 13,000 candidates.

Web Development Intern

Aug 2022 – Sept 2022

Musketeers Tech (Private) Ltd

Lahore, Pakistan

- Contributed to a **Pet Store Management System** using the MERN stack, Docker, and Git/GitHub.

PROJECTS

Skewaze – Business Profitability Analytics Platform | *React, Node.js, TypeScript, PostgreSQL*

- Designed and built the web platform for a financial-intelligence product giving business owners real-time visibility into job-level profitability, including cash-flow forecasting, WIP reporting, and AR aging views.
- Delivered a responsive React frontend and supporting Node.js/TypeScript services on PostgreSQL, translating analytics and BI requirements into a functional user-facing product.

Dialer – VoIP Communication Platform | *React, NestJS, PostgreSQL, Sequelize, Twilio, Socket.io*

- Developed a full-stack VoIP platform (React frontend + NestJS backend) integrating Twilio’s third-party APIs for real-time calling, messaging, voicemail, and CSV contact upload.
- Integrated AI-powered transcription endpoints for real-time speech-to-text and built scalable APIs with NestJS and PostgreSQL.

Sugarcane Monitoring Dashboard | *Next.js, NestJS, TypeORM, PostgreSQL, PostGIS, GeoServer, JWT*

- Built a full-stack geospatial platform with an interactive Next.js dashboard (WMS map layers, circle-wise analytics) over a multi-service NestJS backend with PostGIS spatial queries, serving 60K+ crop samples.
- Implemented RBAC with multi-tenant mill isolation, S3-backed field survey image uploads with EXIF geotag extraction, and Excel/PDF report generation.

Remote Crop Monitoring Dashboard | *Next.js, NestJS, TypeORM, PostgreSQL, PostGIS, GeoServer, JWT*

- Built an interactive Next.js dashboard with WMS/WFS map layers and farm health scoring over a NestJS backend integrating GeoServer, NDVI satellite imagery, and real-time weather APIs.
- Implemented JWT auth, paginated access to 3,000+ farmer records, and automated E2E tests covering the API pipeline.

Centralized Auth | *NestJS, TypeORM, PostgreSQL, JWT, Redis, Swagger, Jest, Docker*

- Built a centralized auth system with JWT token rotation, dashboard-scoped RBAC, multi-tenant partner isolation, API key management, and full audit logging.
- Implemented 65 automated E2E tests with database-backed HTML reporting, plus rate limiting and request correlation tracing for observability.

AI Financial Analyst – LLM Automation Pipeline | *n8n, Docker, Google Gemini, JavaScript, REST APIs*

- Built an automated market-analysis pipeline on self-hosted n8n (Docker): tool nodes call external APIs to ingest RSS feeds, then invoke the Google Gemini LLM to synthesize macroeconomic trends into structured output.

CERTIFICATIONS

Fundamentals of Remote Sensing (NASA)	2025
AWS Academy Microservices and CI/CD Pipeline Builders (AWS)	2024
freeCodeCamp: Back End Development and APIs	2023
freeCodeCamp: JavaScript Algorithms and Data Structures	2022
DataCamp: Introduction to Python	2022